

Transposes, Permutations, Vector Spaces

Finish elimination with $PA = LU$, flip matrices with transposes, then step up from vectors to whole spaces of vectors, and the column space of a matrix.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. Permutations: reordering rules

Re-listing the equations in another order: why a 0 forces it, $PA = LU$, and undoing it with P^T .

2. Transposes

Rows become columns. Portfolio return as $r^T x$, and $(AB)^T = B^T A^T$ entry by entry.

3. Symmetric matrices and $R^T R$

Why $R^T R$ is always symmetric, and how covariance matrices come from it.

4. Vector spaces

A set you can add and scale inside without leaving: portfolios as a space.

5. When a set is not a space

Long-only portfolios, fixed-budget portfolios and unions fail the test.

6. Subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$

Lines and planes through the origin. Zero-cost (self-financing) trades form one.

7. The column space $C(A)$

All combinations of the columns: exactly the targets your funds can reach.

METHODS IN THIS LESSON

[PA = LU](#)

[P⁻¹ = P^T](#)

[Transpose](#)

[Dot product \$x^T y\$](#)

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PART 2 Prerequisites

Click any link to review it, then come back.

- Span and column space, first look (Lesson 1): "every point you can reach by mixing".
- Permutation matrices (Lesson 2): PA swaps rows, AP swaps columns.

- $PA = LU$ and transposes of products (Lesson 4).
- The entry rule and building-block trades (Lesson 3).

PART 3 Explanation

THE RUNNING EXAMPLE: THE SAME FUNDS, NEW QUESTIONS

Funds: Nifty (12%, beta 1.0), bonds (7%, beta 0), midcap (15%, beta 1.2). Rules: budget, return (₹ thousand), market exposure. The fund matrix and the matrix of **scenario returns** we'll use:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \quad R = \begin{bmatrix} -10 & 4 \\ 10 & 7 \\ 30 & 10 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{bad year} \\ \leftarrow \text{normal year} \\ \leftarrow \text{good year} \end{array}$$

R lists the yearly return (%) of Nifty (column 1) and bonds (column 2) in three scenarios. This lesson asks three new questions. What happens when rows and columns swap roles? Which sets of portfolios behave like a "space"? And exactly which targets can a set of funds reach?

1. Permutations: changing the order of the rules

THE WHOLE IDEA IN ONE LINE

A **permutation** changes only the **order in which the equations are listed**. The equations themselves don't change, so **the answer doesn't change**. A permutation matrix P is just the instruction for the new order.

Step 1: two rules, two orders

Our two-fund problem has two rules. We can write them down in either order:

Order A (budget first)	Order B (return first)
(1) budget: $x + y = 10$	(1) return: $12x + 7y = 105$
(2) return: $12x + 7y = 105$	(2) budget: $x + y = 10$

It's the same two facts on a different line of the page, so both lists have the same answer: $x = 7$, $y = 3$ (check: $7 + 3 = 10$ and $84 + 21 = 105$). As matrices, the rows simply trade places:

$$\text{Order A: } \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \longrightarrow \text{Order B: } \begin{bmatrix} 12 & 7 \\ 1 & 1 \end{bmatrix}$$

The matrix that makes this swap is $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. How to make it and how to read it:

1 MAKE IT

Do the same swap to the identity matrix $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$: swapping its two rows gives $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

2 READ ROW 1

$(0, 1)$ means "take 0 of rule 1 and 1 of rule 2": $0 \times (1, 1) + 1 \times (12, 7) = (12, 7)$. The return rule now comes first.

READ ROW 2

$(1, 0)$ means "take rule 1": $1 \times (1, 1) + 0 \times (12, 7) = (1, 1)$. The budget rule now comes second.

3 DON'T FORGET B

The right-hand sides move with their rules: $Pb = P(10, 105) = (105, 10)$.

WHAT WE HAVE NOW

$PA = \begin{bmatrix} 12 & 7 \\ 1 & 1 \end{bmatrix}$ and $Pb = (105, 10)$: Order B, same answer $(7, 3)$.

Step 2: why would anyone reorder? A zero in the way

Take the hedging problem (sell x Nifty futures and y Nifty IT futures) and suppose someone wrote the currency rule first:

$$\begin{array}{ll} y = 3 & \text{(currency)} \\ x + y = 8 & \text{(market)} \end{array} \iff \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

1 ELIMINATION'S FIRST MOVE

Use the top-left number (the first **pivot**) to remove x from the row below. The multiplier is (number below) $\div$ (pivot) = $1 \div 0$.

PROBLEM

You can't divide by 0. The first rule has no x in it, so it can't be used to remove x from anything. Elimination is stuck.

2 FIX

Swap the two rules, using $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ again: market first, then currency.

WHAT WE HAVE NOW

$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}$. It's already triangular: the last rule says $y = 3$, then the first says $x + 3 = 8$, so $x = 5$.

That is the only reason permutations show up in elimination: a 0 lands where a pivot should be, and swapping in a lower row that has a non-zero number there gets things moving again.

Step 3: three rules, six orders: try them

With three rules (market, currency, rates) there are 6 ways to list them. Writing **M**, **C**, **R** for the rules: M-C-R, M-R-C, C-M-R, C-R-M, R-M-C, R-C-M. Pick an order below. You'll see the rules in that order, the permutation matrix P that produces it, and whether elimination runs straight through or gets stuck on a 0.

M-C-R
M-R-C
C-M-R
C-R-M
R-M-C
R-C-M

1. **market:** $x + y + z = 10$
2. **currency:** $y = 3$
3. **rates:** $z = 2$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{row } i \text{ of } P \text{ says which rule goes in place } i)$$

Elimination runs straight through: pivots 1, 1, 1. Back substitution: $z = 2$, $y = 3$, $x = 10 - 3 - 2 = 5$.

Order	First rule listed	Elimination without swaps	Answer
M-C-R	market: $x + y + z = 10$	runs straight through (pivots 1, 1, 1)	(5, 3, 2)
M-R-C	market	stuck at pivot 2: the second rule (rates) has no y	(5, 3, 2) after a swap
C-M-R, C-R-M	currency: $y = 3$	stuck at pivot 1: no x in the first rule	(5, 3, 2) after a swap
R-M-C, R-C-M	rates: $z = 2$	stuck at pivot 1: no x in the first rule	(5, 3, 2) after a swap

Two lessons from the table. **The answer is (5, 3, 2) in every order:** reordering never changes it. And only **M-C-R** works without help, because the market rule is the only one containing x (Nifty futures), so it has to come first.

Step 4: $PA = LU$: reorder first, then eliminate

Elimination with a swap in the middle is untidy. Strang's cleaner way: **do all the reordering first**, then eliminate normally. The reordering is P ; the normal elimination gives L and U (Lesson 4).

RULE · PA = LU

For every invertible A there is a permutation P that puts the rules in a workable order; then $PA = LU$ with no more swaps. If the order was already fine, $P = I$ and it's just $A = LU$.

One worked case: the credit fund. Lesson 2's credit fund earns 12% like Nifty but has beta 0. Rules in the usual order (budget, return, exposure):

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 12 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}, \quad b = \begin{bmatrix} 10 \\ 126 \\ 7.4 \end{bmatrix}$$

1 WHERE IT GETS STUCK

Return $- 12 \times$ budget gives $(12 - 12, 12 - 12, 15 - 12) = (0, 0, 3)$. Nifty **and** the credit fund disappear together (both earn 12), so row 2 has a 0 where the second pivot should be.

THE FIX

List the exposure rule second instead: order budget, exposure, return. That's P_{23} (swap rows 2 and 3 of I): $P_{23}A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1.2 \\ 12 & 12 & 15 \end{bmatrix}$.

2 NOW ELIMINATE

Exposure $- 1 \times$ budget: $(1 - 1, 0 - 1, 1.2 - 1) = (0, -1, 0.2)$.

Return $- 12 \times$ budget: $(0, 0, 3)$. The entry below the second pivot is already 0. Done: $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0.2 \\ 0 & 0 & 3 \end{bmatrix}$, and the multipliers 1, 12, 0 give

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 12 & 0 & 1 \end{bmatrix}.$$

3 CHECK LU = PA

Row 2 of LU : $1 \times (1, 1, 1) + 1 \times (0, -1, 0.2) = (1, 0, 1.2) \checkmark$. Row 3: $12 \times (1, 1, 1) + 1 \times (0, 0, 3) = (12, 12, 15) \checkmark$. (Row 1 is just $(1, 1, 1) \checkmark$.)

4 SOLVE

Reorder b the same way: $Pb = (10, 7.4, 126)$. Forward: $c = (10, 7.4 - 10, 126 - 120) = (10, -2.6, 6)$. Back: $3z = 6$ so $z = 2$; $-y + 0.4 = -2.6$ so $y = 3$; $x + 3 + 2 = 10$ so $x = 5$.

ANSWER

₹5 lakh Nifty, ₹3 lakh credit, ₹2 lakh midcap. Check: $10 \checkmark, 60 + 36 + 30 = 126 \checkmark, 5 + 2.4 = 7.4 \checkmark$.

Step 5: undoing a reorder: $P^{-1} = P^T$

To undo a reorder, put the rules back. Two easy cases:

- A **swap**: swapping twice puts everything back, so $P \times P = I$. Check for $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: rows of PP are $0(0, 1) + 1(1, 0) = (1, 0)$ and $1(0, 1) + 0(1, 0) = (0, 1)$, which is I ✓. A swap is its own undo.
- A **cycle**: P turns the order (M, C, R) into (C, R, M). To undo it, turn (C, R, M) back into (M, C, R). The matrix that does that turns out to be the transpose P^T .

RULE · $P^{-1} = P^T$

The undo of any reorder is its transpose: $P^T P = I$. So you never need to "compute" the inverse of a permutation: just flip it.

Check for the cycle. $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ (take rule 2, rule 3, rule 1) and $P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Multiply $P^T P$ row by row: row 1 of P^T is (0, 0, 1), "take row 3 of P " = (1, 0, 0); row 2 is (1, 0, 0), "take row 1 of P " = (0, 1, 0); row 3 is (0, 1, 0), "take row 2 of P " = (0, 0, 1). That is I ✓.

How many orders are there? 2 rules: 2 orders. 3 rules: $3 \times 2 \times 1 = 6$ (any of 3 first, either of the other 2 next, the last one is forced). n rules: $n! = n(n-1) \cdots 1$, which is 24 for 4 rules and 120 for 5.

▼ For the curious: small pivots, and permutations as a "family"

Why software swaps more than it has to. Algebra only needs a swap when a pivot is exactly 0. Programs like MATLAB also swap when a pivot is merely *small*, because a tiny pivot means a huge multiplier, which magnifies rounding errors. Example: $0.0001x + y = 1$ and $x + y = 2$ (true answer $x \approx 1.0001$, $y \approx 0.9999$). With pivot 0.0001 the multiplier is 10,000; if y is rounded to 1.000, then $x = (1 - y)/0.0001 = 0$, completely wrong. Swap first and the pivot is 1: the same rounded y gives $x = 2 - y = 1.000$, correct. Picking the largest available pivot is called **partial pivoting**.

A closed family. Reordering twice is still a reordering, so multiplying two permutation matrices gives another one, and every one has an undo. For example, swap rows 2 and 3, then rows 1 and 2: $P_{12}P_{23} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, a cycle. Mathematicians call such a family a *group*.

IN ONE LINE

A permutation only re-lists the rules (and their right-hand sides); it's needed when a 0 sits where a pivot should be; $PA = LU$ means "reorder first, then eliminate"; and the undo is P^T .

▼ Check yourself (MIT problem 5.1): find a 3×3 permutation P , not the identity, with $P^3 = I$.

The cycle $(M, C, R) \rightarrow (C, R, M)$. Apply it three times: $(C, R, M) \rightarrow (R, M, C) \rightarrow (M, C, R)$, back to the start. So $P^3 = I$.

2. Transposes

METHOD · TRANSPOSE

The transpose A^T flips A across its main diagonal: rows become columns and columns become rows. In symbols, $(A^T)_{ij} = A_{ji}$: the entry in row i , column j of A^T is the entry in row j , column i of A . An $m \times n$ matrix has an $n \times m$ transpose.

Transposing the fund matrices

1 2×2

$A = \begin{bmatrix} 1 & 12 \\ 12 & 7 \end{bmatrix}$. Row 1 $(1, 12)$ becomes column 1; row 2 $(12, 7)$ becomes column 2: $A^T = \begin{bmatrix} 1 & 12 \\ 12 & 7 \end{bmatrix}$.

CHECK ONE ENTRY

$(A^T)_{12}$ should be $A_{21} = 12$ ✓.

2 3×3

RESULT

Each row of A becomes a column:

$A^T = \begin{bmatrix} 1 & 12 & 1 \\ 12 & 7 & 0 \\ 1 & 15 & 1.2 \end{bmatrix}$. Diagonal entries 1, 7, 1.2 stay put; $A_{21} = 12$ moved to position $(1, 2)$, $A_{32} = 0$ to $(2, 3)$, and so on.

3 RECTANGULAR

MEANING

With only Nifty and bonds but all three rules, A is 3×2 : $\begin{bmatrix} 1 & 12 \\ 12 & 7 \\ 1 & 0 \end{bmatrix}$. Its transpose is 2×3 : $\begin{bmatrix} 1 & 12 & 1 \\ 12 & 7 & 0 \end{bmatrix}$.

In A each **column** is a fund; in A^T each **row** is a fund's full profile: Nifty = $(1, 12, 1)$, bonds = $(1, 7, 0)$.

Row vectors and the dot product

Vectors are columns. Their transposes are rows. That gives a compact way to write a dot product:

METHOD · DOT PRODUCT AS $X^T Y$

For column vectors x and y with n entries, $x^T y$ is a $(1 \times n)(n \times 1) = 1 \times 1$ product: a single number, $x_1 y_1 + \cdots + x_n y_n$.

Portfolio return in one line. Let $r = (12, 7, 15)$ be the returns per lakh and $x = (5, 3, 2)$ the holdings. Then

$$r^T x = \begin{bmatrix} 12 & 7 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} = 12 \times 5 + 7 \times 3 + 15 \times 2 = 60 + 21 + 30 = 111.$$

₹111 thousand: exactly the return rule (row 2 of A) applied to client 1. Finance writes portfolio return as $w^T r$ ("weights transpose returns") all the time; this is that formula.

The transpose of a product reverses the order

RULE · $(AB)^T = B^T A^T$

Transpose each factor and reverse the order, just like inverses (Lesson 4). Also $(A^T)^T = A$: flipping twice returns the original.

Check it on Lesson 3's two clients, $A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$ and $X = \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix}$, where $AX = \begin{bmatrix} 10 & 105 \\ 105 & 90 \end{bmatrix}$:

1 LEFT SIDE

Transpose AX : $(AX)^T = \begin{bmatrix} 10 & 105 \\ 105 & 90 \end{bmatrix}$.

2 RIGHT SIDE

$X^T = \begin{bmatrix} 7 & 3 \\ 4 & 6 \end{bmatrix}$ and $A^T = \begin{bmatrix} 1 & 12 \\ 1 & 7 \end{bmatrix}$. Entry by entry: $(7 \cdot 1 + 3 \cdot 1, 7 \cdot 12 + 3 \cdot 7) = (10, 105)$ for row 1, and $(4 \cdot 1 + 6 \cdot 1, 4 \cdot 12 + 6 \cdot 7) = (10, 90)$ for row 2.

WHAT WE HAVE NOW $X^T A^T = \begin{bmatrix} 10 & 105 \\ 105 & 90 \end{bmatrix} = (AX)^T \checkmark$.

3 MEANING

AX had one **column** per client; $(AX)^T$ has one **row** per client: (budget, return) for client 1, then client 2. That's the same report sorted the other way, like a spreadsheet with clients as rows.

WHY NOT $A^T X^T$? $A^T X^T = \begin{bmatrix} 1 \cdot 7 + 12 \cdot 4 & 1 \cdot 3 + 12 \cdot 6 \\ 1 \cdot 7 + 7 \cdot 4 & 1 \cdot 3 + 7 \cdot 6 \end{bmatrix} = \begin{bmatrix} 55 & 75 \\ 35 & 45 \end{bmatrix}$: wrong order, meaningless numbers.

IN ONE LINE

$(A^T)_{ij} = A_{ji}$; $x^T y$ is a dot product; $(AB)^T = B^T A^T$.

▼ Check yourself (Strang's example): transpose $\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$.

Column 1 (1, 2, 4) becomes row 1 and column 2 (3, 3, 1) becomes row 2: $\begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}$, a 2×3 matrix.

3. Symmetric matrices and $R^T R$

DEFINITION: SYMMETRIC MATRIX

A square matrix is **symmetric** if transposing doesn't change it: $S^T = S$, that is $S_{ij} = S_{ji}$ for every i, j . Everything above the diagonal is mirrored below it. Example: $\begin{bmatrix} 3 & 1 & 7 \\ 1 & 2 & 9 \\ 7 & 9 & 4 \end{bmatrix}$.

Strang: symmetric matrices "show up in a lot of applications", and he shows where they come from. Take **any** matrix R , even a rectangular one, and multiply its transpose by it.

RULE · $R^T R$ IS ALWAYS SYMMETRIC

For any $m \times n$ matrix R , the $n \times n$ matrix $R^T R$ is symmetric. Proof in two lines: $(R^T R)^T = R^T (R^T)^T = R^T R$, using "reverse the order" and $(R^T)^T = R$. (RR^T , which is $m \times m$, is symmetric for the same reason.)

$R^T R$ for the scenario returns, entry by entry

R is 3×2 (3 scenarios $\times$ 2 funds), so $R^T R$ is 2×2 . Entry (i, j) = (column i of R) $\cdot$ (column j of R). The columns are Nifty $(-10, 10, 30)$ and bonds $(4, 7, 10)$.

1 ENTRY (1, 1)

$$\text{Nifty} \cdot \text{Nifty} = (-10)^2 + 10^2 + 30^2 = 100 + 100 + 900 = 1100.$$

2 ENTRY (1, 2)

$$\text{Nifty} \cdot \text{bonds} = (-10)(4) + (10)(7) + (30)(10) = -40 + 70 + 300 = 330.$$

ENTRY (2, 1)

bonds $\cdot$ Nifty: the same three products in the other order, $4(-10) + 7(10) + 10(30) = 330$. **This is the symmetry:** a dot product doesn't care which vector comes first.

3 ENTRY (2, 2)

$$\text{bonds} \cdot \text{bonds} = 4^2 + 7^2 + 10^2 = 16 + 49 + 100 = 165.$$

WHAT WE HAVE NOW $R^T R = \begin{bmatrix} 1100 & 330 \\ 330 & 165 \end{bmatrix}$: symmetric ✓.

The 3×3 version RR^T compares **scenarios** instead of funds (entry $(i, j) = \text{row } i \cdot \text{row } j \text{ of } R$):

$$RR^T = \begin{bmatrix} (-10)^2 + 4^2 & (-10)(10) + 4 \cdot 7 & (-10)(30) + 4 \cdot 10 \\ \cdot & 10^2 + 7^2 & 10 \cdot 30 + 7 \cdot 10 \\ \cdot & \cdot & 30^2 + 10^2 \end{bmatrix} = \begin{bmatrix} 116 & -72 & -260 \\ -72 & 149 & 370 \\ -260 & 370 & 1000 \end{bmatrix}$$

Only the upper triangle needed computing; symmetry fills in the rest.

MODEL · COVARIANCE MATRIX (PREVIEW)

Subtract each fund's average return first, then form $\frac{1}{m}C^TC$ (with m scenarios): that is the **covariance matrix** of the funds, the core input of portfolio risk models. Because it has the form C^TC , a covariance matrix is always symmetric.

1 AVERAGES

Nifty: $(-10 + 10 + 30)/3 = 10$. Bonds: $(4 + 7 + 10)/3 = 7$.

SUBTRACT THEM

Nifty: $(-10 - 10, 10 - 10, 30 - 10) = (-20, 0, 20)$. Bonds: $(4 - 7, 7 - 7, 10 - 7) = (-3, 0, 3)$. These are the columns of C .

2 C^TC

$(1, 1)$: $400 + 0 + 400 = 800$. $(1, 2) = (2, 1)$: $(-20)(-3) + 0 + 20 \cdot 3 = 60 + 60 = 120$. $(2, 2)$: $9 + 0 + 9 = 18$.

DIVIDE BY 3

Covariance = $\begin{bmatrix} 800/3 & 40 \\ 40 & 6 \end{bmatrix} \approx \begin{bmatrix} 266.7 & 40 \\ 40 & 6 \end{bmatrix}$.

MEANING

Nifty's variance ≈ 266.7 (volatility $\approx \sqrt{266.7} \approx 16.3\%$), bonds' variance 6 ($\approx 2.4\%$), and a positive covariance 40: in these scenarios the two funds move together. Risk models come back in the probability course.

IN ONE LINE

$S^T = S$ means mirrored across the diagonal; R^TR is always symmetric, which is why covariance matrices are.

▼ Check yourself (MIT problem 5.2): how many entries of a 4×4 matrix can be chosen freely if it is symmetric? If it is skew-symmetric ($A^T = -A$)?

Symmetric: the 4 diagonal entries plus the 6 above the diagonal (the 6 below are copies): **10**.

Skew-symmetric: the diagonal must be 0 (since $a_{ii} = -a_{ii}$) and the lower triangle is minus the upper: **6**.

4. Vector spaces

Strang calls this "the beginning of real linear algebra": instead of single vectors, we look at whole **spaces** of vectors. What do we do with vectors? Two things: **add** them and **multiply them by numbers**. A space is a collection where those two operations never take you outside it.

RULE · VECTOR SPACE TEST (CLOSURE)

A collection V of vectors is a **vector space** if, for any v, w in V and any real number c : (1) $v + w$ is in V (closed under addition), and (2) $c v$ is in V (closed under multiplication by scalars). Together: every **linear combination** $c v + d w$ stays in V . (There are eight more rules about how addition behaves, such as $v + w = w + v$; for vectors of numbers they hold automatically.)

The big examples: $\mathbb{R}^n$

$\mathbb{R}^2$ is all vectors with 2 real components: the whole xy -plane. In portfolio terms, think of (x, y) as "₹ lakh in Nifty, ₹ lakh in bonds", where negative numbers mean selling short. Check the two rules on real portfolios:

1 ADD

Client 1 $(7, 3)$ plus client 2 $(4, 6)$, component by component: $(7 + 4, 3 + 6) = (11, 9)$. Merging two books gives another pair of numbers: still in $\mathbb{R}^2$ ✓.

2 SCALE

Double client 1: $2(7, 3) = (14, 6)$. Reverse it: $-1 \times (7, 3) = (-7, -3)$ (short both). Scale by zero: $0 \times (7, 3) = (0, 0)$, the empty portfolio. All still in $\mathbb{R}^2$ ✓.

$\mathbb{R}^3$ is all vectors with 3 components, for example three-fund portfolios $(5, 3, 2)$, and $\mathbb{R}^n$ is all vectors with n components. Careful: $(3, 2, 0)$ is in $\mathbb{R}^3$, not $\mathbb{R}^2$. It has three components; one of them happens to be zero.

WATCH OUT: THE ZERO VECTOR IS ESSENTIAL

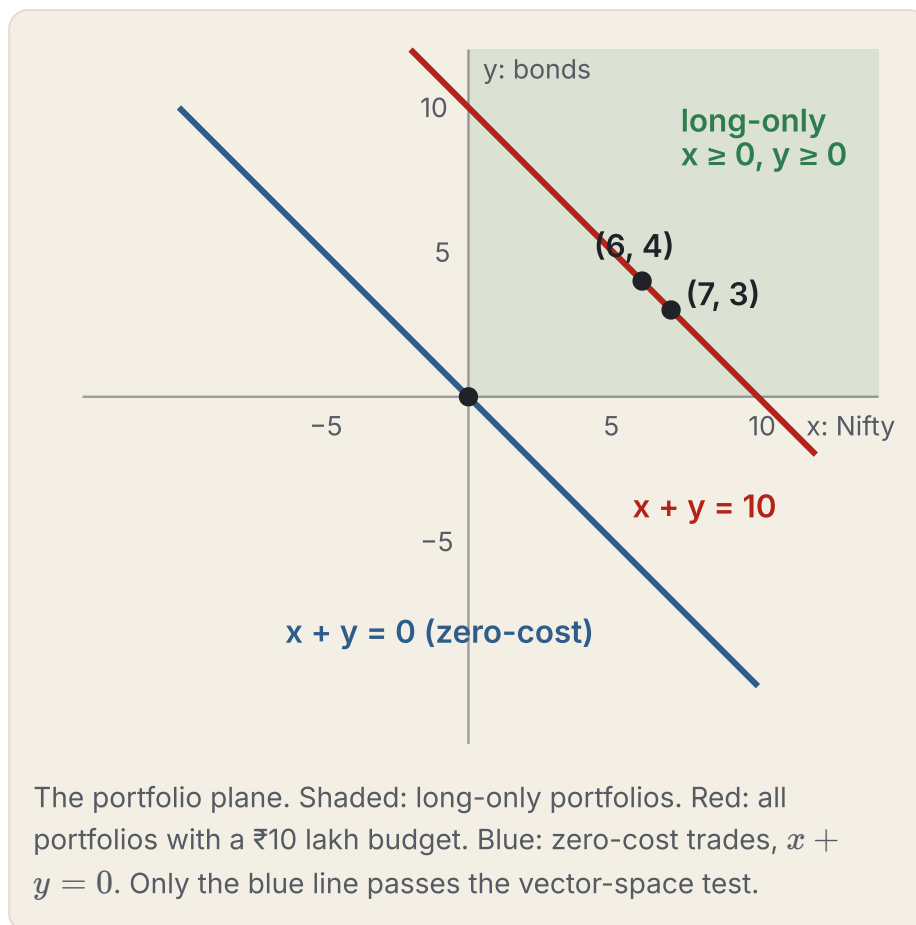
Every vector space contains 0 , because you must be allowed to multiply any vector by the number 0 . Strang: "Every vector space has got that zero vector in it." A plane with the origin removed is **not** a space.

IN ONE LINE

A vector space is closed under adding and scaling, so under all linear combinations, and always contains 0.

5. When a set is not a space

The test is easiest to understand from sets that fail it. Here are three, all natural in portfolio work.



Failure 1: long-only portfolios (a quadrant)

Many funds may only **buy**, never sell short: $x \geq 0$ and $y \geq 0$. Strang uses exactly this quarter-plane as his first non-example.

1 ADDITION

Two long-only portfolios, $(7, 3) + (6, 4) = (13, 7)$: both components still ≥ 0 ✓. Adding is fine.

2 SCALING

Multiply $(7, 3)$ by -1 : $(-7, -3)$, both negative, so outside the quadrant ✗. (Strang multiplies by -5 ; any negative number fails.)

VERDICT

Closed under addition, **not** under scalar multiplication: **not a vector space**.

IN MARKETS

This is why a **long-only** constraint changes the maths of portfolio construction. Without it, the set of allowed portfolios is a vector space and linear algebra solves everything directly. With it, you need inequality-constrained optimisation, a different toolkit.

Failure 2: a fixed budget (a line not through the origin)

All portfolios that use exactly ₹10 lakh: $x + y = 10$ (the red line).

1 ADDITION

$(7, 3)$ and $(6, 4)$ both have budget 10. Their sum $(13, 7)$ has budget $13 + 7 = 20 \neq 10$ ✗.

2 ZERO

$0 \times (7, 3) = (0, 0)$ has budget $0 \neq 10$ ✗. The zero vector is not on the line.

VERDICT

Not a vector space. Strang: a line in $\mathbb{R}^2$ that misses the origin is never a subspace.

Failure 3: "all-Nifty or all-bonds" (a union of two lines)

From the recitation: take every all-Nifty portfolio $(t, 0)$ together with every all-bond portfolio $(0, t)$.

1 SCALING

A multiple of an all-Nifty portfolio is still all-Nifty, and the same for bonds ✓.

2 ADDITION

$(1, 0) + (0, 1) = (1, 1)$: a mixed portfolio, on neither line ✗.

VERDICT

Not a vector space. Each line alone is a subspace, but their union isn't: adding escapes it.

IN ONE LINE

To fail, a set only has to let one sum or one multiple escape: long-only (negatives), fixed budget (sums and 0), unions (sums).

6. Subspaces of $\mathbb{R}^2$ and $\mathbb{R}^3$

RULE · SUBSPACE TEST

A **subspace** is a vector space sitting inside a bigger one. To check that a set inside $\mathbb{R}^n$ is a subspace: (1) it contains 0; (2) sums stay inside; (3) multiples stay inside. Strang: once a subspace contains a non-zero vector v , it must contain the **whole line** of multiples cv .

All the subspaces of $\mathbb{R}^2$

Subspace	Example in portfolio terms
All of $\mathbb{R}^2$	every portfolio of Nifty and bonds, long or short
A line through $(0, 0)$	all-Nifty portfolios $(t, 0)$; or zero-cost trades $(t, -t)$
Just $\{(0, 0)\}$	the empty portfolio alone

There is nothing in between. A line through the origin inside $\mathbb{R}^2$ looks like $\mathbb{R}^1$ but isn't the same thing: its vectors still have **two** components (Strang's point).

Zero-cost trades: a subspace that matters

MODEL · SELF-FINANCING (ZERO-COST) TRADES

A trade is **self-financing** (zero-cost) if the money it puts into some funds is exactly taken out of others: the changes add up to zero. Rebalancing an existing portfolio without adding or withdrawing money is always a zero-cost trade.

With two funds the zero-cost trades are $\{(x, y) : x + y = 0\}$, the blue line through $(1, -1)$. Run the subspace test:

1 ZERO

$(0, 0): 0 + 0 = 0 \checkmark$ ("do nothing" costs nothing).

2 SUMS

Two zero-cost trades $(a, -a)$ and $(b, -b)$ add to $(a + b, -a - b) = (a + b, -(a + b))$: components still add to 0 $\checkmark$. Example: $(2, -2) + (1, -1) = (3, -3)$.

3 MULTIPLES

$c(a, -a) = (ca, -ca)$: still adds to 0 $\checkmark$. Example: $-3 \times (2, -2) = (-6, 6)$.

VERDICT

A **subspace**: a line through the origin. Compare with "budget = 10": the only difference is the right-hand side, 0 instead of 10, and that 0 is what lets the origin in.

All the subspaces of $\mathbb{R}^3$

Subspace	Example with three funds
All of $\mathbb{R}^3$	every three-fund portfolio
A plane through 0	zero-cost trades: $x + y + z = 0$
A line through 0	zero-cost trades that also keep market exposure unchanged
Just $\{0\}$	the empty trade

Working out that line. A trade (x, y, z) that changes neither the budget nor the exposure needs $x + y + z = 0$ and $x + 1.2z = 0$.

1

**FROM THE EXPOSURE
RULE
FROM THE BUDGET
RULE**

$$x = -1.2z.$$

$$y = -x - z = 1.2z - z = 0.2z.$$

2**SO**

every such trade is $z \times (-1.2, 0.2, 1)$. Taking $z = 5$ gives whole numbers: $(-6, 1, 5)$.

CHECK

Budget: $-6 + 1 + 5 = 0$ ✓. Exposure: $-6 + 1.2 \times 5 = -6 + 6 = 0$ ✓. Return: $12(-6) + 7(1) + 15(5) = -72 + 7 + 75 = 10$.

MEANING

The trades that change **only the return** form a line, and this one adds ₹10 thousand. It is exactly $10 \times$ the building-block column $(-0.6, 0.1, 0.5)$ of A^{-1} in Lesson 3.

That line is where the zero-cost plane and the zero-exposure plane meet. In general, **the intersection of two subspaces is a subspace** (both conditions survive), while a union usually isn't (topic 5). Add a third condition, "zero return change", and only $(0, 0, 0)$ is left: the three planes meet only at the origin, because the fund matrix is invertible.

IN ONE LINE

Subspaces are $\{0\}$, lines and planes through the origin, and the whole space. Zero-cost trades are a subspace; fixed-budget portfolios are not.

▼ **Check yourself (MIT problem 5.3):** do the symmetric 2×2 matrices form a subspace of all 2×2 matrices? Do the non-symmetric ones?

Symmetric: yes. If $A^T = A$ and $B^T = B$ then $(A + B)^T = A^T + B^T = A + B$ and $(cA)^T = cA^T = cA$; and the zero matrix is symmetric. Non-symmetric: no. $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, which is symmetric, so the sum escapes (and 0 isn't in the set at all).

7. The column space $C(A)$

Now the subspace that comes straight out of a matrix. Strang: take the columns, and "not satisfied with a few vectors, we want a space of vectors".

METHOD · COLUMN SPACE $C(A)$

The **column space** $C(A)$ of an $m \times n$ matrix is the set of **all linear combinations of its columns**. It is a subspace of $\mathbb{R}^m$: combinations of combinations are still combinations, and $0 \times$ every column gives 0.

Two funds, three rules: a plane in $\mathbb{R}^3$

Use only Nifty and bonds, but keep all three rules. Then A is 3×2 and its columns live in $\mathbb{R}^3$:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \\ 1 & 0 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{array} \quad C(A) = \left\{ x \begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} \right\}$$

Each choice of (x, y) gives one reachable target. All of them together fill a **plane through the origin** in $\mathbb{R}^3$: more than a line (the columns point in different directions), but not all of $\mathbb{R}^3$ (only two directions).

Which targets are in it?

Look at the exposure row: $1 \cdot x + 0 \cdot y = E$. With only Nifty (beta 1) and bonds (beta 0), **the exposure always equals the Nifty amount**. That one fact decides everything:

1 TARGET

$(10, 105, 7)$: ₹10 lakh, ₹105 thousand, exposure 7.

HOW

Exposure row: $x = 7$. Budget row: $7 + y = 10$, so $y = 3$. Return row (the check): $12(7) + 7(3) = 84 + 21 = 105 \checkmark$.

VERDICT

In $C(A)$: $7 \times (1, 12, 1) + 3 \times (1, 7, 0) = (10, 105, 7)$.

2 TARGET

HOW

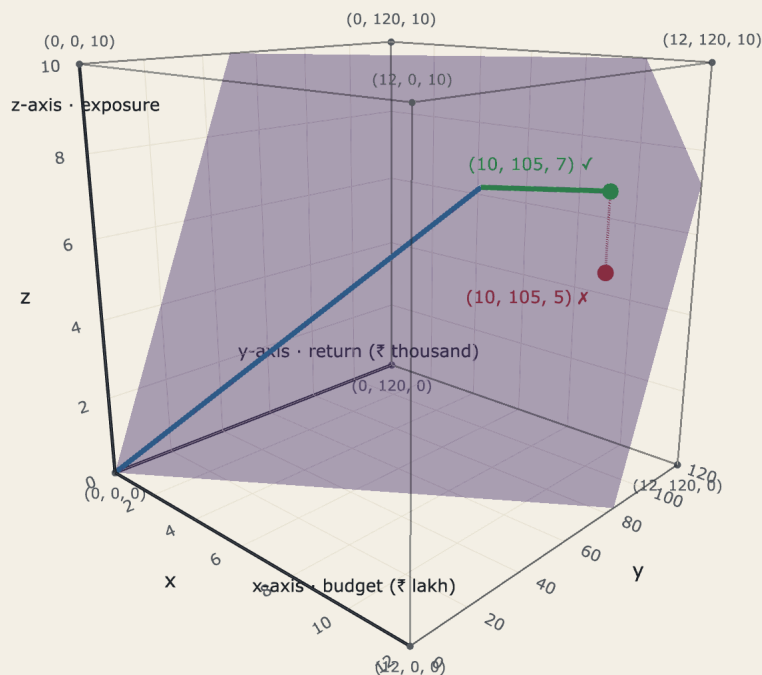
VERDICT

$(10, 105, 5)$: the same budget and return, exposure 5.

Exposure row: $x = 5$. Budget row: $y = 5$. Return row: $12(5) + 7(5) = 60 + 35 = 95 \neq 105$.

Not in $C(A)$: no mix of Nifty and bonds gives ₹105 thousand with exposure only 5. (You would need a third fund, such as midcap.)

The plane's equation. Every reachable (B, R, E) satisfies one equation: $-7B + R - 5E = 0$. Check it on both columns: Nifty $-7 + 12 - 5 = 0$ ✓, bonds $-7 + 7 - 0 = 0$ ✓. Then the targets: $(10, 105, 7)$ gives $-70 + 105 - 35 = 0$ (in), while $(10, 105, 5)$ gives $-70 + 105 - 25 = 10$ (out by 10). (The numbers $(-7, 1, -5)$ come from the *cross product* of the two columns, a 3D tool from the calculus course.)



- $C(A)$: the plane $-7B + R - 5E = 0$
 $7 \times$ Nifty $(1, 12, 1)$
 $+ 3 \times$ bonds $(1, 7, 0)$
- $(10, 105, 7) \checkmark$ in $C(A)$
 $(10, 105, 5) \times$ not in $C(A)$
 gap: exposure must be 7

The column space $C(A)$: a plane through the origin (drag to rotate). 7 Nifty steps plus 3 bond steps reach $(10, 105, 7)$ on the plane. The target $(10, 105, 5)$ sits off it: no mix of these two funds reaches it.

Three funds: the whole space

Add the midcap column $(1, 15, 1.2)$. The three columns point in three independent directions (the matrix is invertible, Lesson 3), so $C(A)$ is **all of** $\mathbb{R}^3$: every (budget, return, exposure) target is reachable, which is why every client so far had an answer. With the balanced fund instead, the three columns lie in one plane and $C(A)$ is only that plane (Lesson 1).

WHAT'S NEXT

$Ax = b$ has a solution **exactly when b is in $C(A)$** . The next lesson turns this into the full story of when, and how many, solutions exist.

IN ONE LINE

$C(A)$ = all combinations of the columns = every target the funds can reach; it is a line, a plane or the whole space.

▼ **Check yourself (Strang's example):** what is the column space of $\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$? Is $(4, 5, 5)$ in it?

The plane through the origin containing $(1, 2, 4)$ and $(3, 3, 1)$ (they point in different directions, so it is a plane, not a line). And $(4, 5, 5) = 1 \times (1, 2, 4) + 1 \times (3, 3, 1)$, so yes: it is the sum of the two columns.